Block Matrices and Determinants

Linear Algebra · Lecture 5

Last time (Lecture 4, Definition 10, Theorem 12(d)): $\operatorname{rank} A$ is the number of nonzero rows of an RREF of A; $A \in \mathbb{R}^{n\times n}$ is invertible $\iff$ $\operatorname{rank} A = n$. Throughout, e_j is column j of I; row_i, col_j: row i, column j; $A(i|j)$: A without row i and column j.

From Lecture 1 and Lecture 2 (restated).

- Lecture 1, Definitions 1, 2, Theorems 3, 4: $A = B \iff$ all entries agree; $(A+B)_{ij} = a_{ij} + b_{ij}$, $(cA)_{ij} = ca_{ij}$; $(AB)_{ij} = \sum_r a_{ir}b_{rj}$; $(AB)C = A(BC)$, $A(B+C) = AB+AC$, $(A+B)C = AC+BC$, $c(AB) = (cA)B = A(cB)$, $I_mA = A = AI_n$, $0A = 0$, $A0 = 0$; a 1×1 factor c acts as a number: $McN = c(MN)$. For $A = [a_1\ \cdots\ a_n]$, $B = [b_1\ \cdots\ b_p]$: $\operatorname{col}_j(AB) = Ab_j$, $Ax = \sum_j x_j a_j$, $\operatorname{row}_i(AB) = \operatorname{row}_i(A)B$, $AB = \sum_l a_l \operatorname{row}_l(B)$ (**outer product expansion**).
- Lecture 1, Definitions 7, 12: $(A')_{ij} = a_{ji}$; A is **symmetric** if $A' = A$. $\operatorname{diag}(d_1,\dots,d_n)$: diagonal entries d_i, other entries 0; $A \in \mathbb{R}^{n\times n}$ is **upper (lower) triangular** if $a_{ij} = 0$ for $i > j$ ($i < j$).
- Lecture 2, Definitions 2, 5: row operations $R_r \leftarrow cR_r$ ($c \neq 0$), $R_r \leftarrow R_r + cR_s$, $R_r \leftrightarrow R_s$ ($r \neq s$). Forward elimination (Steps 1–4; Step 3, scaling a pivot to 1, may be postponed) gives leading entries in columns $k_1 < k_2 < \cdots$, zero rows at the bottom; backward elimination (Step 5) adds multiples of each pivot row to the rows above it, to clear the entries above the pivot.
- Lecture 2, Definitions 10, 13, Theorems 11(a), 14(a),(c), 15, Proposition 17: A is **invertible** if $AB = BA = I$ for some B, the unique inverse A^{-1}; $(A^{-1})^{-1} = A$. The **elementary matrices** $D_r(c)$ ($c \neq 0$), $T_{rs}(c)$, P_{rs} are I after $R_r \leftarrow cR_r$, $R_r \leftarrow R_r + cR_s$, $R_r \leftrightarrow R_s$; a row operation ω is $\omega(A) = EA$, $E = \omega(I)$; if $\omega_1,\dots,\omega_k$ take A to B, then $B = E_k\cdots E_1A$. $A \in \mathbb{R}^{n\times n}$ is invertible $\iff$ an RREF of A is I_n $\iff$ A is a product of elementary matrices; $BA = I$ or $AB = I \Rightarrow B = A^{-1}$. $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)^{-1} = \frac{1}{ad-bc}\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$ if $ad - bc \neq 0$.

1. Block matrices

Definition 1. A **partition** of $A \in \mathbb{R}^{m\times n}$ splits the rows into consecutive groups of $m_1,\dots,m_r$ rows and the columns into consecutive groups of $n_1,\dots,n_s$ columns. With $M_0 = N_0 = 0$, $M_i = m_1+\cdots+m_i$, $N_j = n_1+\cdots+n_j$, the **block** $A_{ij} \in \mathbb{R}^{m_i\times n_j}$ has the entries $(A_{ij})_{xy} = a_{M_{i-1}+x,\,N_{j-1}+y}$; $A = (A_{ij})$ is the array of its blocks. Matrices of the same size, partitioned alike, are equal $\iff$ their blocks are equal. A partition of $B \in \mathbb{R}^{m\times n}$ is **conformal for addition** if it has the same groups; one of $B \in \mathbb{R}^{n\times p}$ is **conformal for multiplication** (in AB) if its row groups are $n_1,\dots,n_s$ (its column groups $p_1,\dots,p_t$ are free).

Theorem 2. Let $A \in \mathbb{R}^{m\times n}$ be partitioned into blocks $A_{ij} \in \mathbb{R}^{m_i\times n_j}$ ($1 \leqslant i \leqslant r$, $1 \leqslant j \leqslant s$).

(a) If $B \in \mathbb{R}^{m\times n}$ is partitioned conformally for addition, into blocks B_{ij}, then $A+B$ has the blocks $A_{ij}+B_{ij}$, and cA the blocks cA_{ij} ($c \in \mathbb{R}$).

(b) A', with row groups $n_1,\dots,n_s$ and column groups $m_1,\dots,m_r$, has the (j,i) block A'_{ij}.

(c) If $B \in \mathbb{R}^{n\times p}$ is partitioned conformally for multiplication, into blocks $B_{jk} \in \mathbb{R}^{n_j\times p_k}$, then AB, with row groups $m_1,\dots,m_r$ and column groups $p_1,\dots,p_t$, has the blocks $(AB)_{ik} = A_{i1}B_{1k} + \cdots + A_{is}B_{sk} \in \mathbb{R}^{m_i\times p_k}$.

Proof. (a) For $u = M_{i-1}+x$, $v = N_{j-1}+y$: $(A+B)_{uv} = a_{uv}+b_{uv} = (A_{ij}+B_{ij})_{xy}$, $(cA)_{uv} = (cA_{ij})_{xy}$.
(b) Row group j of A': rows $N_{j-1}+1,\dots,N_j$; column group i: columns $M_{i-1}+1,\dots,M_i$ (Lecture 1, Definition 7):

$$(A')_{N_{j-1}+y,\,M_{i-1}+x} = a_{M_{i-1}+x,\,N_{j-1}+y} = (A_{ij})_{xy} = (A'_{ij})_{yx}$$

(c)

- $A_{i*} \in \mathbb{R}^{m_i \times n}$: the rows of A in row group i, $A_{i*} = [A_{i1} \ \cdots \ A_{is}]$; $B_{*k} \in \mathbb{R}^{n \times p_k}$: the columns of B in column group k, the blocks $B_{1k}, \ldots, B_{sk}$ stacked (conformality).
- The columns of AB in group k are those of AB_{*k} ($\mathrm{col}_j(AB) = Ab_j$); the rows of AB_{*k} in group i are those of $A_{i*}B_{*k}$ ($\mathrm{row}_i(AB) = \mathrm{row}_i(A)B$) (Lecture 1, Theorem 4(a),(c)). So $(AB)_{ik} = A_{i*}B_{*k}$.
- Inner sum: $A_{i*}B_{*k} = \sum_{l=1}^{n} \mathrm{col}_l(A_{i*})\,\mathrm{row}_l(B_{*k})$ by the outer product expansion (Lecture 1, Theorem 4(d)), and the indices $N_{j-1} < l \leqslant N_j$ give $\sum_{y=1}^{n_j} \mathrm{col}_y(A_{ij})\,\mathrm{row}_y(B_{jk}) = A_{ij}B_{jk}$; adding the s stretches, $(AB)_{ik} = \sum_j A_{ij}B_{jk}$. □

For $r = s = t = 2$:

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}$$

$\mathrm{diag}(A_1, \ldots, A_r)$ has the square diagonal blocks A_i and zero blocks elsewhere; by (c), with the same groups, $\mathrm{diag}(A_1, \ldots, A_r)\,\mathrm{diag}(B_1, \ldots, B_r) = \mathrm{diag}(A_1B_1, \ldots, A_rB_r)$ (only $j = i = k$ gives a nonzero term).

2. Block elimination and the Schur complement

Theorem 3. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p \times p}$ invertible, $B \in \mathbb{R}^{p \times q}$, $C \in \mathbb{R}^{n \times p}$, $D \in \mathbb{R}^{n \times q}$, and $S = D - CA^{-1}B \in \mathbb{R}^{n \times q}$, the **Schur complement** of A in M. Then

$$\begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_n \end{pmatrix} M = \begin{pmatrix} A & B \\ 0 & S \end{pmatrix}, \qquad M \begin{pmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} A & 0 \\ C & S \end{pmatrix}$$

Proof.

Left factor: column groups (p, n), the row groups of M (Theorem 2(c)):

$$\begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_n \end{pmatrix} M = \begin{pmatrix} I_pA + 0C & I_pB + 0D \\ -CA^{-1}A + I_nC & -CA^{-1}B + I_nD \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}$$

Right factor: row groups (p, q), the column groups of M (Theorem 2(c)):

$$M \begin{pmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} AI_p + B0 & -AA^{-1}B + BI_q \\ CI_p + D0 & -CA^{-1}B + DI_q \end{pmatrix} = \begin{pmatrix} A & 0 \\ C & D - CA^{-1}B \end{pmatrix}$$

□

When D is invertible, $A - BD^{-1}C$ is likewise the **Schur complement** of D in M.

Corollary 4. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p \times p}$, $D \in \mathbb{R}^{q \times q}$, and let A and $S = D - CA^{-1}B$ be invertible. Then M is invertible and

$$M^{-1} = \begin{pmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{pmatrix} \begin{pmatrix} A^{-1} & 0 \\ 0 & S^{-1} \end{pmatrix} \begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{pmatrix} = \begin{pmatrix} A^{-1} + A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{pmatrix}$$

Proof.

E_3, E_2, E_1: the three factors, from left to right; Theorem 2(c):

$$E_1M = \begin{pmatrix} A & B \\ 0 & S \end{pmatrix}, \quad E_2E_1M = \begin{pmatrix} A^{-1}A & A^{-1}B \\ S^{-1} \cdot 0 & S^{-1}S \end{pmatrix} = \begin{pmatrix} I_p & A^{-1}B \\ 0 & I_q \end{pmatrix} \quad \text{(Theorem 3, } n = q\text{)}$$

$$E_3E_2E_1M = \begin{pmatrix} I_p & A^{-1}B - A^{-1}B \\ 0 & I_q \end{pmatrix} = I \implies M^{-1} = E_3E_2E_1 \qquad \text{(Lecture 2, Theorem 15)}$$

Multiplying out (Theorem 2(c)):

$$E_2E_1 = \begin{pmatrix} A^{-1} & 0 \\ -S^{-1}CA^{-1} & S^{-1} \end{pmatrix}, \qquad E_3(E_2E_1) = \begin{pmatrix} A^{-1}+A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{pmatrix}$$

□

On $[M \mid I]$ with block rows R_1, R_2, left multiplication by E_1, E_2, E_3 is the **block row operation** $R_2 \leftarrow R_2 - CA^{-1}R_1$, then $R_1 \leftarrow A^{-1}R_1$ and $R_2 \leftarrow S^{-1}R_2$, then $R_1 \leftarrow R_1 - A^{-1}BR_2$ (Theorem 2(c)); together they give $[I \mid M^{-1}]$: **block elimination**.

Example 5. Find the Schur complement S of A, and M^{-1}, for $M \in \mathbb{R}^{5\times 5}$ with groups $(2,3)$ for rows and columns:

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \end{pmatrix}, \quad C = B', \quad D = \begin{pmatrix} 3 & 2 & 3 \\ 2 & 3 & 3 \\ 3 & 3 & 7 \end{pmatrix}$$

Solution.

A^{-1} (Lecture 2, Proposition 17: $2\cdot 1 - 1\cdot 1 = 1$); $A^{-1}B$, CA^{-1}; S (Theorem 3):

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \qquad A^{-1}B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad CA^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$$

$$S = D - CA^{-1}B = \begin{pmatrix} 3 & 2 & 3 \\ 2 & 3 & 3 \\ 3 & 3 & 7 \end{pmatrix} - \begin{pmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \\ 3 & 2 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

S^{-1} ($SS^{-1} = I_3$; Lecture 2, Theorem 15); the off-diagonal blocks of M^{-1} (Corollary 4):

$$S^{-1} = \begin{pmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}, \qquad -A^{-1}BS^{-1} = \begin{pmatrix} -4 & 3 & -2 \\ 1 & -1 & 0 \end{pmatrix}, \qquad -S^{-1}CA^{-1} = \begin{pmatrix} -4 & 1 \\ 3 & -1 \\ -2 & 0 \end{pmatrix}$$

The upper left block $A^{-1}+A^{-1}BS^{-1}CA^{-1} = A^{-1} - (A^{-1}B)(-S^{-1}CA^{-1})$; M^{-1} (Corollary 4); a check of row 1:

$$\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}\begin{pmatrix} -4 & 1 \\ 3 & -1 \\ -2 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} - \begin{pmatrix} -6 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 7 & -2 \\ -2 & 3 \end{pmatrix}$$

$$M^{-1} = \left(\begin{array}{cc|ccc} 7 & -2 & -4 & 3 & -2 \\ -2 & 3 & 1 & -1 & 0 \\ \hline -4 & 1 & 3 & -2 & 1 \\ 3 & -1 & -2 & 2 & -1 \\ -2 & 0 & 1 & -1 & 1 \end{array}\right),$$

$$\begin{aligned} \mathrm{row}_1(M)M^{-1} &= 2(7,-2,-4,3,-2) + (-2,3,1,-1,0) \\ &\quad + 2(-4,1,3,-2,1) + (3,-1,-2,2,-1) \\ &\quad + 3(-2,0,1,-1,1) \\ &= (1,0,0,0,0) \end{aligned}$$

Theorem 6. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p\times p}$, $D \in \mathbb{R}^{q\times q}$, and let D and $F = A - BD^{-1}C$ be invertible. Then M is invertible and

$$M^{-1} = \begin{pmatrix} F^{-1} & -F^{-1}BD^{-1} \\ -D^{-1}CF^{-1} & D^{-1} + D^{-1}CF^{-1}BD^{-1} \end{pmatrix}$$

Proof. For $X = \left(\begin{smallmatrix} P & Q \\ R & T \end{smallmatrix}\right)$, by Theorem 2(c) and Definition 1, $MX = I \iff AP + BR = I_p$, $CP + DR = 0$, $AQ + BT = 0$, $CQ + DT = I_q$.

Solve the four equations:

$$\begin{aligned} &R = -D^{-1}CP \implies (A - BD^{-1}C)P = FP = I_p \implies P = F^{-1},\ R = -D^{-1}CF^{-1} \\ &T = D^{-1} - D^{-1}CQ \implies AQ + BT = FQ + BD^{-1} = 0 \\ &\implies Q = -F^{-1}BD^{-1},\ T = D^{-1} + D^{-1}CF^{-1}BD^{-1} \end{aligned}$$

Conversely, these blocks satisfy the four equations:

$$AP + BR = (A - BD^{-1}C)F^{-1} = I_p, \qquad CP + DR = CF^{-1} - CF^{-1} = 0$$
$$AQ + BT = -(A - BD^{-1}C)F^{-1}BD^{-1} + BD^{-1} = 0$$
$$CQ + DT = -CF^{-1}BD^{-1} + I_q + CF^{-1}BD^{-1} = I_q$$
$$MX = I \implies M^{-1} = X \qquad \text{(Lecture 2, Theorem 15)}$$

□

3. Determinants

Definition 7. For $A \in \mathbb{R}^{n\times n}$, the **determinant** $\det A$ is defined by induction on n: $\det(a) = a$ for $n = 1$; for $n \geqslant 2$, with the **minors** $\det A(i|j)$ and the **cofactors** $c_{ij} = (-1)^{i+j} \det A(i|j)$,

$$\det A = a_{11}c_{11} + a_{21}c_{21} + \cdots + a_{n1}c_{n1} \qquad \text{(expansion along column 1)}$$

For $1 \leqslant j \leqslant n$ let $E_j(A) = a_{1j}c_{1j} + \cdots + a_{nj}c_{nj}$ (along column j), and $E_1(a) = a$ for $n = 1$. So $E_1(A) = \det A$.

Theorem 8. Let $n \geqslant 1$, $1 \leqslant j \leqslant n$, $A \in \mathbb{R}^{n\times n}$, rows $r \neq s$, and $c \in \mathbb{R}$.

(a) E_j is linear in each row: multiplying row r by c multiplies $E_j(A)$ by c; if A_1, A_2, A_3 agree except in row r and $\mathrm{row}_r(A_3) = \mathrm{row}_r(A_1) + \mathrm{row}_r(A_2)$, then $E_j(A_3) = E_j(A_1) + E_j(A_2)$.

(b) Adding c times row s to row r does not change $E_j(A)$.

(c) Interchanging rows r and s changes the sign of $E_j(A)$. So $E_j(A) = 0$ if two rows of A are equal.

(d) $E_j(I_n) = 1$.

Proof. Induction on n. For $n = 1$, $E_1(a) = a$ satisfies (a) and (d); (b), (c) concern two rows. Let $n \geqslant 2$, assume (a)–(d) for $\det = E_1$ on $\mathbb{R}^{(n-1)\times(n-1)}$, and write $E = E_j$, $E(A) = \sum_i (-1)^{i+j} a_{ij} \det A(i|j)$.

- (a): $i \neq r$: a_{ij} does not involve row r, and $\det A(i|j)$ is linear in it ($A(i|j)$ contains row r without entry j; induction); $i = r$: $\det A(r|j)$ does not involve row r, and a_{rj} is linear in it.
- Rows k, $k+1$ equal $\Rightarrow E(A) = 0$: $\det A(i|j) = 0$ for $i \neq k, k+1$ (two equal rows; induction, (c)); the terms $i = k, k+1$ cancel, as $a_{kj} = a_{k+1,j}$, $A(k|j) = A(k+1|j)$ and the signs are $(-1)^{k+j}$, $(-1)^{k+1+j}$.
- Adjacent interchange: with $E(u, v)$ the value when rows k, $k+1$ are u', v' ($u, v \in \mathbb{R}^n$), $0 = E(u+v, u+v) = E(u,u) + E(u,v) + E(v,u) + E(v,v) = E(u,v) + E(v,u)$ ((a), previous point).
- (c): interchanging rows $r < s$ is $2(s-r) - 1$ adjacent interchanges (row s up to position r, then the former row r down to position s), an odd number. Two equal rows: interchanging them, $E(A) = -E(A)$.
- (b): by (a) in row r, the new value is $E(A) + cE(A_s)$, $A_s = A$ with row r replaced by row s; $E(A_s) = 0$ by (c).
- (d): $\mathrm{col}_j(I_n) = e_j$, $I_n(j|j) = I_{n-1}$: $E(I_n) = (-1)^{2j} \det I_{n-1} = 1$ (induction). □

Theorem 9. Let $A \in \mathbb{R}^{n\times n}$ and $1 \leqslant j \leqslant n$.

(a) If A has a zero row, then $E_j(A) = 0$.

(b) If A is upper triangular, then $E_j(A) = a_{11}a_{22}\cdots a_{nn}$; in particular $E_j(\operatorname{diag}(d_1,\ldots,d_n)) = d_1\cdots d_n$.

(c) $E_j(D_r(c)) = c$, $E_j(T_{rs}(c)) = 1$, $E_j(P_{rs}) = -1$, and $E_j(FA) = E_j(F)E_j(A)$ for every elementary matrix $F \in \mathbb{R}^{n\times n}$.

Proof.

(a) Multiplying the zero row by 2 does not change A (Theorem 8(a)):

$$E_j(A) = 2E_j(A) \implies E_j(A) = 0$$

(b)

- All $a_{ii} \neq 0$: $R_i \leftarrow a_{ii}^{-1}R_i$ (all i) gives B, upper triangular with diagonal entries 1, $E_j(B) = E_j(A)/(a_{11}\cdots a_{nn})$ (Theorem 8(a)); backward elimination (Lecture 2, Definition 5) with these pivots turns B into I_n by additions: $E_j(B) = E_j(I_n) = 1$ (Theorem 8(b),(d)).
- Some $a_{ii} = 0$, i the largest: row i is 0 in columns $1,\ldots,i$. For $l = i+1,\ldots,n$ in turn, subtract a multiple of row l from row i to clear entry (i,l) ($a_{ll} \neq 0$; row l is 0 before column l, so the zeros made stay). This keeps $E_j(A)$ (Theorem 8(b)) and makes row i zero: $E_j(A) = 0 = a_{11}\cdots a_{nn}$ by (a).

(c) $D_r(c)$, $T_{rs}(c)$, P_{rs} are I_n after $R_r \leftarrow cR_r$, $R_r \leftarrow R_r + cR_s$, $R_r \leftrightarrow R_s$ (Lecture 2, Definition 13):
$E_j(D_r(c)) = cE_j(I_n) = c$, $E_j(T_{rs}(c)) = E_j(I_n) = 1$, $E_j(P_{rs}) = -E_j(I_n) = -1$ (Theorem 8).
FA is A after the operation of F (Lecture 2, Theorem 14(a)), so $E_j(FA)$ is $cE_j(A)$, $E_j(A)$ or $-E_j(A)$ (Theorem 8): $E_j(FA) = E_j(F)E_j(A)$.

□

Theorem 10. Let interchanges of two rows and additions of a multiple of one row to another take $A \in \mathbb{R}^{n\times n}$ to an upper triangular U, with t interchanges. Then $E_j(A) = (-1)^t u_{11}\cdots u_{nn}$ for every j. Forward elimination (Steps 1, 2, 4, without scaling) always ends with such a U. Hence $E_j(A) = \det A$ for every j:

$$\det A = a_{1j}c_{1j} + a_{2j}c_{2j} + \cdots + a_{nj}c_{nj} \qquad (\textbf{expansion along column } j)$$

Proof.

Each addition keeps E_j, each interchange changes its sign (Theorem 8(b),(c)):
$u_{11}\cdots u_{nn} = E_j(U) = (-1)^t E_j(A)$ (Theorem 9(b)).
Forward elimination: leading entries in columns $k_1 < k_2 < \cdots$, so $k_i \geqslant i$, and zero rows at the bottom (Lecture 2, Definition 5): U is upper triangular.
So $E_j(A) = (-1)^t u_{11}\cdots u_{nn} = E_1(A) = \det A$ for every j.

□

If a row is multiplied by $c \neq 0$ on the way, det is multiplied by c (Theorem 8(a)): divide by c at the end.

Example 11. Compute $\det A$ for

$$A = \begin{pmatrix} 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 0 \\ 2 & 6 & 4 & 4 \\ 1 & 0 & 2 & 3 \end{pmatrix}$$

Solution.

$a_{11} = 0$: begin with an interchange; then additions (Theorem 10):

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 2 & 6 & 4 & 4 \\ 1 & 0 & 2 & 3 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 2 & 2 & 4 \\ 0 & -2 & 1 & 3 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 2 & 4 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - 2R_3} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & -2 \end{pmatrix} = U, \quad t = 1 : \ \det A = -\Big(1 \cdot 2 \cdot 1 \cdot (-2)\Big) = 4$$

Theorem 12. Let $A, B \in \mathbb{R}^{n \times n}$.

(a) A is invertible $\iff \det A \neq 0 \iff \operatorname{rank} A = n$.

(b) $\det(AB) = \det A \det B$.

(c) If A is invertible, $\det A^{-1} = 1/\det A$. Moreover $\det(cA) = c^n \det A$ ($c \in \mathbb{R}$) and $\det A^k = (\det A)^k$ ($k = 1, 2, \ldots$).

Proof.

(a) An RREF $R = F_k \cdots F_1 A$ of A (Lecture 2, Theorem 14(c)); $\operatorname{rank} A$ and invertibility: Lecture 4, Definition 10, Theorem 12(d); $\det I_n = 1$ (Theorem 8(d)):

$$\begin{aligned} \det R = \det F_k \cdots \det F_1 \det A, \quad \det F_i \in \{c \neq 0,\ 1,\ -1\} \quad & \text{(Theorem 9(c))} \\ A \text{ invertible} \implies R = I_n \implies \det A \neq 0 \quad & \text{(Lecture 2, Theorem 15)} \\ A \text{ not invertible} \iff \operatorname{rank} A < n \implies R \text{ has a zero row} \quad & \text{(Lecture 4)} \\ \implies \det R = 0 \implies \det A = 0 \quad & \text{(Theorem 9(a))} \end{aligned}$$

(b) Two cases (Lecture 2, Theorem 15):

A not invertible: then AB is not invertible (an inverse G of AB would give $A(BG) = I$), and $\det(AB) = 0 = \det A \det B$ by (a).

A invertible: $A = F_1 \cdots F_k$ with elementary F_i, and $\det(AB) = \det F_1 \cdots \det F_k \det B$ (Theorem 9(c)); with $B = I_n$, $\det A = \det F_1 \cdots \det F_k$; so $\det(AB) = \det A \det B$.

(c) From (b); powers: $A^{k+1} = A^k A$ (Lecture 1, after Theorem 3):

$$\begin{aligned} \det A \det A^{-1} = \det(AA^{-1}) = \det I_n = 1 & \\ \det(cA) = \det(cI_n) \det A = c^n \det A \quad & \text{(Theorem 9(b))} \\ \det A^{k+1} = \det(A^k A) = \det A^k \det A \quad & \text{(induction on } k\text{)} \end{aligned}$$

□

Theorem 13. Let $A \in \mathbb{R}^{n\times n}$.

(a) If $n \geqslant 2$: $\det A = a_{i1}c_{i1} + a_{i2}c_{i2} + \cdots + a_{in}c_{in}$ for every i (**expansion along row** i).

(b) $\det A' = \det A$.

Proof.

(a) $\operatorname{row}_i(A) = \sum_j a_{ij}e'_j$; $G_j = A$ with row i replaced by e'_j; $H_j = G_j$ after $R_k \leftarrow R_k - a_{kj}R_i$ $(k \neq i)$, so $\operatorname{col}_j(H_j) = e_i$, $H_j(i|j) = A(i|j)$:

$$\begin{aligned} \det A &= \textstyle\sum_j a_{ij} \det G_j && \text{(Theorem 8(a))} \\ \det G_j &= \det H_j = (-1)^{i+j} \det A(i|j) = c_{ij} && \text{(Theorems 8(b), 10, column } j\text{)} \end{aligned}$$

(b) Induction on n; $n = 1$: $A' = A$. For $n \geqslant 2$, $A'(k|1) = A(1|k)'$:

$$\begin{aligned} \det A' &= \textstyle\sum_k (-1)^{k+1} a_{1k} \det A(1|k)' && \text{(Definition 7)} \\ &= \textstyle\sum_k (-1)^{1+k} a_{1k} \det A(1|k) = \det A && \text{(induction; (a), } i = 1\text{)} \end{aligned}$$

□

An operation on the columns of A is the same operation on the rows of A'; so Theorem 8(a)–(c) and Theorem 9(a) hold for columns, and a lower triangular A has $\det A = \det A' = a_{11}\cdots a_{nn}$ (Theorem 9(b)). For $n = 2$, $\det \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = ad - bc$ (Definition 7), the number of Lecture 2, Proposition 17.

4. The adjugate and Cramer's rule

Theorem 14. Let $n \geqslant 2$ and $A \in \mathbb{R}^{n\times n}$ with cofactors c_{ij}. The **adjugate** of A is $\operatorname{adj} A = (c_{ij})'$, that is, $(\operatorname{adj} A)_{ij} = c_{ji}$. Then $A \operatorname{adj} A = (\det A) I_n$; if $\det A \neq 0$, then $A^{-1} = \frac{1}{\det A} \operatorname{adj} A$. If A is symmetric, then $c_{ji} = c_{ij}$.

Proof.

Entry (i, k) of $A \operatorname{adj} A$; $\tilde{A} = A$ with row k replaced by row i $(k \neq i)$:

$$\begin{aligned} (A \operatorname{adj} A)_{ik} &= \textstyle\sum_j a_{ij}c_{kj} = \det A \ (k = i) && \text{(Theorem 13(a), row } i\text{)} \\ \tilde{A}(k|j) = A(k|j) &\implies \textstyle\sum_j a_{ij}c_{kj} = \det \tilde{A} = 0 \ (k \neq i) && \text{(Theorem 13(a), row } k\text{; Theorem 8(c))} \\ A\Big(\tfrac{1}{\det A} \operatorname{adj} A\Big) = I_n &\implies A^{-1} = \tfrac{1}{\det A} \operatorname{adj} A && \text{(Lecture 2, Theorem 15)} \end{aligned}$$

A symmetric: deleting row j and column i of $A = A'$:

$$A(j|i) = A'(j|i) = A(i|j)' \implies \det A(j|i) = \det A(i|j) \implies c_{ji} = c_{ij} \qquad \text{(Theorem 13(b))}$$

□

Example 15. Find A^{-1} by Theorem 14 for $A = \left(\begin{smallmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ -1 & 0 & 0 & 1 \end{smallmatrix}\right)$.

Solution.

Each 3×3 minor is triangular (Theorem 9(b) and the lines after Theorem 13), or has a row with one nonzero entry (Theorem 13(a)):

$$\begin{aligned} c_{11} &= \det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{12} &= -\det \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{13} &= \det \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 1 \end{smallmatrix}\right) = -1, & c_{14} &= -\det \left(\begin{smallmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{smallmatrix}\right) = 1 \\ c_{21} &= -\det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right) = -1, & c_{22} &= \det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{23} &= -\det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{24} &= \det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{smallmatrix}\right) = -1 \\ c_{31} &= \det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{32} &= -\det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{smallmatrix}\right) = -1, & c_{33} &= \det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{34} &= -\det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 0 \end{smallmatrix}\right) = 1 \\ c_{41} &= -\det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) = -1, & c_{42} &= \det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) = 1, & c_{43} &= -\det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = -1, & c_{44} &= \det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 1 \end{aligned}$$

$\det A$ along row 1 (Theorem 13(a)); A^{-1} (Theorem 14):

$$\det A = 1\cdot c_{11} + 1\cdot c_{12} + 0\cdot c_{13} + 0\cdot c_{14} = 2, \qquad A^{-1} = \frac{1}{2}\operatorname{adj} A = \frac{1}{2}\begin{pmatrix} 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 1 \end{pmatrix}$$

Theorem 16. Let $A \in \mathbb{R}^{n\times n}$ ($n \geqslant 2$) with $\det A \neq 0$, $b \in \mathbb{R}^n$, and $A_j = A$ with column j replaced by b. Then $Ax = b$ has exactly one solution x, and $x_j = \det A_j / \det A$ ($1 \leqslant j \leqslant n$) (**Cramer's rule**).

Proof.
$A^{-1} = \frac{1}{\det A}\operatorname{adj} A$ (Theorem 14); $A_j(i|j) = A(i|j)$, entry (i,j) of A_j is b_i:

$$\begin{aligned} Ax = b &\implies x = A^{-1}(Ax) = A^{-1}b; \quad A(A^{-1}b) = b \\ &\implies \text{the only solution is } x = \tfrac{1}{\det A}(\operatorname{adj} A)b \\ \Big((\operatorname{adj} A)b\Big)_j &= \textstyle\sum_i (\operatorname{adj} A)_{ji} b_i = \sum_i c_{ij} b_i \\ \det A_j &= \textstyle\sum_i b_i (-1)^{i+j} \det A(i|j) = \sum_i c_{ij} b_i \qquad \text{(Theorem 10, column } j\text{)} \\ x_j &= \det A_j / \det A \end{aligned}$$

□

Example 17. For A of Example 15 and $b = (3,1,0,0)'$, find x_3 in $Ax = b$ by Cramer's rule, and check it against $x = A^{-1}b$.

Solution.
$\det A = 2$; row 3 of A_3 has one nonzero entry, at $(3,4)$; expand along it, then along row 3 (Theorem 13(a)); Theorem 16; $(\operatorname{adj} A)b$ by Lecture 1, Theorem 4(b):

$$\det A_3 = \det\begin{pmatrix} 1 & 1 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 \end{pmatrix} = (-1)^{3+4}\det\begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ -1 & 0 & 0 \end{pmatrix} = -(-1)(-1)^{3+1}\det\begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} = -2$$

$$x_3 = \frac{\det A_3}{\det A} = \frac{-2}{2} = -1$$

$$x = \tfrac{1}{2}(\operatorname{adj} A)b = \tfrac{1}{2}\Big(3\operatorname{col}_1(\operatorname{adj} A) + \operatorname{col}_2(\operatorname{adj} A)\Big) = \tfrac{1}{2}\Big(3(1,1,-1,1)' + (-1,1,1,-1)'\Big) = (1,2,-1,1)'$$

5. Determinants of block matrices

Theorem 18. Let $A \in \mathbb{R}^{p\times p}$, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{q\times p}$, $D \in \mathbb{R}^{q\times q}$. Then $\det \left(\begin{smallmatrix} A & B \\ 0 & D \end{smallmatrix}\right) = \det A \det D = \det \left(\begin{smallmatrix} A & 0 \\ C & D \end{smallmatrix}\right)$; in particular $\det \operatorname{diag}(A, D) = \det A \det D$.

Proof.
Three factors: expand p times along row 1, q times along the last row (Theorem 13(a)); Theorem 9(b):

$$\det\begin{pmatrix} I_p & 0 \\ 0 & D \end{pmatrix} = \det D, \qquad \det\begin{pmatrix} A & 0 \\ 0 & I_q \end{pmatrix} = \det A, \qquad \det\begin{pmatrix} I_p & B \\ 0 & I_q \end{pmatrix} = 1$$

The product (Theorem 2(c)); Theorem 12(b); the transpose (Theorem 2(b), Theorem 13(b)):

$$\begin{pmatrix} I_p & 0 \\ 0 & D \end{pmatrix}\begin{pmatrix} I_p & B \\ 0 & I_q \end{pmatrix}\begin{pmatrix} A & 0 \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} I_p & 0 \\ 0 & D \end{pmatrix}\begin{pmatrix} A & B \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} \implies \det\begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det D \cdot 1 \cdot \det A$$

$$\det\begin{pmatrix} A & 0 \\ C & D \end{pmatrix} = \det\begin{pmatrix} A' & C' \\ 0 & D' \end{pmatrix} = \det A' \det D' = \det A \det D$$

□

Theorem 19. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p\times p}$, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{q\times p}$, $D \in \mathbb{R}^{q\times q}$.

(a) If A is invertible, then $\det M = \det A \det(D - CA^{-1}B)$.

(b) If D is invertible, then $\det M = \det D \det(A - BD^{-1}C)$.

Proof.

(a) The left factor of Theorem 3 ($n = q$) is lower triangular with diagonal entries 1 (the lines after Theorem 13):

$$1 \cdot \det M = \det \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix} = \det A \det(D - CA^{-1}B) \qquad \text{(Theorems 12(b), 18)}$$

(b) The left factor is upper triangular with diagonal entries 1 (Theorems 2(c), 9(b)):

$$\begin{pmatrix} I_p & -BD^{-1} \\ 0 & I_q \end{pmatrix} M = \begin{pmatrix} A - BD^{-1}C & B - BD^{-1}D \\ C & D \end{pmatrix} = \begin{pmatrix} A - BD^{-1}C & 0 \\ C & D \end{pmatrix}$$

$1 \cdot \det M = \det(A - BD^{-1}C) \det D$ (Theorems 12(b), 18).

□

If A is invertible, $\det A \neq 0$, so M is invertible $\iff$ $D - CA^{-1}B$ is (by (a) and Theorem 12(a)).

Theorem 20. Let $A \in \mathbb{R}^{n\times n}$ and $T \in \mathbb{R}^{m\times m}$ be invertible, $U \in \mathbb{R}^{n\times m}$ and $V \in \mathbb{R}^{m\times n}$.

(a) $\det(A+UTV) = \det A \det T \det(T^{-1}+VA^{-1}U)$. Hence $A+UTV$ is invertible $\iff$ $T^{-1}+VA^{-1}U$ is invertible, and then (the **Sherman–Morrison–Woodbury formula**)

$$(A + UTV)^{-1} = A^{-1} - A^{-1}U(T^{-1} + VA^{-1}U)^{-1}VA^{-1}$$

(b) For $u, v \in \mathbb{R}^n$, $A + uv'$ is invertible $\iff$ $1 + v'A^{-1}u \neq 0$, and then $(A + uv')^{-1} = A^{-1} - \frac{1}{1+v'A^{-1}u}A^{-1}uv'A^{-1}$.

Proof. (a) Let $M = \left(\begin{smallmatrix} A & -U \\ V & T^{-1} \end{smallmatrix}\right)$. As $(T^{-1})^{-1} = T$ (Lecture 2, Theorem 11(a)), the Schur complements of A and of T^{-1} in M are $E = T^{-1} - VA^{-1}(-U) = T^{-1} + VA^{-1}U$ and $F = A - (-U)TV = A + UTV$.

Determinants:

$$\det M = \det A \det E = \det T^{-1} \det F \qquad \text{(Theorem 19(a),(b))}$$
$$\det F = \det A \det T \det E, \quad \det A, \det T \neq 0 \qquad \text{(Theorem 12(c),(a))}$$
$$F \text{ invertible} \iff \det F \neq 0 \iff \det E \neq 0 \iff E \text{ invertible} \qquad \text{(Theorem 12(a))}$$

E, F invertible: the upper left block of M^{-1}, with the blocks A, $-U$, V, T^{-1} of M:

$$(M^{-1})_{11} = A^{-1} + A^{-1}(-U)E^{-1}VA^{-1} \qquad \text{(Corollary 4, Schur complement } E \text{ of } A)$$
$$(M^{-1})_{11} = F^{-1} \qquad \text{(Theorem 6, Schur complement } F \text{ of } T^{-1})$$
$$F^{-1} = A^{-1} - A^{-1}UE^{-1}VA^{-1} \qquad (M^{-1} \text{ is unique: Lecture 2, Definition 10})$$

(b) In (a), $m = 1$, $U = u$, $T = (1)$, $V = v'$:
$A+UTV = A+uv'$ and $T^{-1}+VA^{-1}U = (1+v'A^{-1}u)$, a 1×1 matrix: invertible $\iff$ $1+v'A^{-1}u \neq 0$ (Theorem 12(a), $\det(a) = a$), with inverse $\big(1/(1 + v'A^{-1}u)\big)$ (Lecture 2, Definition 10). As a 1×1 factor acts as a number (Lecture 1, Theorem 3(b)),

$$A^{-1}u\Big(1/(1 + v'A^{-1}u)\Big)v'A^{-1} = \frac{1}{1 + v'A^{-1}u}A^{-1}uv'A^{-1}$$

□

Practice. The Exercises and Problems of the textbook, Chapter 5: block products and transposes, block elimination and the Schur complement, determinants by row reduction and by cofactor expansion, the product rule, the adjugate and Cramer's rule, determinants of block matrices, and the Sherman–Morrison–Woodbury formula.